φ^4 in Doi–Peliti, expanded in a Hermite basis

Concrete construction of the "wild forest" — and how CFAC tames it

(Reply to Gunnar's aside in his 1 May 2026 note)

Abstract

Gunnar's aside: *"can analytic combinatorics help characterising the wild forest of diagrams you get when you attempt to write down φ^4-theory in Doi–Peliti using a Hermite basis?"* This note answers concretely. We compute the four-vertex tensor $V_{n_1n_2n_3n_4} = \int h_{n_1}h_{n_2}h_{n_3}h_{n_4}\,dx$ explicitly for $n_i \in \{0,1,2,3\}$ via `hermite_vertices.py`, list the surviving 1-loop self-energy diagrams for external modes 0 and 1, draw them, contrast with plane-wave φ^4, and write out the multivariate Lagrange functional equation that generates the entire forest from one polynomial. The selection rule is parity ($\sum n_i$ even) — not the triangle inequality our concept sketch claimed. That correction matters: it determines how aggressively CFAC's projections collapse the count.

1 Why a Hermite basis at all? What's at stake?

Plane-wave φ^4 assumes translation invariance. In any setting where the field sees a confining potential or a defect — harmonic trap, soft wall, defect line, optical lattice, a particle bound to a slow background — translation invariance is broken and plane waves stop being the natural one-particle basis. The eigenstates of $-\frac{1}{2}\partial_x^2 + \frac{1}{2}\omega^2 x^2$ are Hermite functions, and any φ^4 on top of that quadratic free theory expands as

$$\varphi(x,t) \;=\; \sum_{n\geq 0} a_n(t)\,h_n(x), \qquad h_n(x) = \frac{H_n(x)\,e^{-x^2/2}}{\sqrt{2^n n!\sqrt{\pi}}}, \qquad \int h_a h_b\,dx = \delta_{ab}.$$

Each Hermite mode then carries its own oscillator energy $\mu_n = (n+\frac{1}{2})\hbar\omega$ and the φ^4 interaction couples the modes through the four-overlap

$$\boxed{V_{n_1n_2n_3n_4} \;\equiv\; \int_{\mathbb{R}} h_{n_1}(x)\,h_{n_2}(x)\,h_{n_3}(x)\,h_{n_4}(x)\,dx \;=\; \Big(\prod_i \frac{1}{\sqrt{2^{n_i}n_i!\sqrt{\pi}}}\Big)\int H_{n_1}H_{n_2}H_{n_3}H_{n_4}\,e^{-2x^2}dx.}$$

The four $\exp(-x^2/2)$ factors from the four h's combine into e^{-2x^2}. We use physicists' Hermite H_n throughout (orthogonal w.r.t. e^{-x^2}).

What the change of basis does to the diagrammatics. Plane-wave φ^4 has *one* vertex (translation-invariant, $g\,\varphi^4$); momentum is conserved at each vertex; 1-loop self-energy is one bubble with one momentum integral. Hermite-basis φ^4 has a vertex tensor $V_{n_1n_2n_3n_4}$ with infinitely many entries (one per quadruple of mode indices), no momentum conservation per se but a parity selection rule, and the 1-loop self-energy of an external mode n is a *sum* of bubbles indexed by the internal mode m, each with coefficient V_{nnmm} and its own loop integral L_m.

Doi–Peliti convention. Below we write Doi–Peliti φ^4 in the form $\widetilde{a}_{n_1}a_{n_2}a_{n_3}a_{n_4}\,V_{n_1n_2n_3n_4}$, treating a and $\widetilde{a}$ as distinct (one creation, one annihilation; propagator is directed, $\widetilde{a}$-leg $\to$ a-leg). For the symmetry-factor calculation in this note we take the four-vertex symmetric in all four legs (the V-tensor is symmetric by construction), and the propagator is the bare oscillator one $\langle a_n\widetilde{a}_{n'}\rangle_0(\omega) = \delta_{nn'}/(-i\omega+\mu_n)$. This is the standard "reaction-diffusion in a trap" setting; see e.g. Cardy's lectures. Where conventions differ (real φ^4 versus DP), the only change is that the real-field propagator is undirected and the Wick-contraction count picks up a global factor of 2 on each internal line; the structure of the V-tensor and the parity selection rule are convention-independent.

2 The four-vertex tensor: explicit values

2.1 Selection rule (parity, only)

$H_n(-x) = (-1)^n H_n(x)$, so the integrand of V has parity $(-1)^{n_1+n_2+n_3+n_4}$. The integration domain is $(-\infty, \infty)$, hence

$$\boxed{V_{n_1n_2n_3n_4} = 0 \quad \text{unless} \quad n_1 + n_2 + n_3 + n_4 \in 2\mathbb{Z}.}$$

Important correction. Our concept sketch (`reply.tex` §Hermite-basis aside) claimed an additional triangle inequality $|n_1 - n_2 - n_3| \le n_4 \le n_1 + n_2 + n_3$. *That is wrong for the four-point overlap.* The triangle inequality does apply to the *three*-point overlap $\int h_a h_b h_c \, dx$ — specifically, Bailey's identity [1] for $\int H_a H_b H_c \, e^{-x^2} dx$ forces both parity ($a+b+c$ even) and $|a-b| \le c \le a+b$. But the four-point overlap is a contraction

$$V_{abcd} = \sum_{e\ge 0} g_{abe}\, g_{cde}, \qquad g_{abc} = \int h_a h_b h_c \, dx,$$

and the sum over the intermediate e broadens the support: as long as one e exists with both $|a-b| \le e \le a+b$ and $|c-d| \le e \le c+d$ and parity-compatible with each pair, $V_{abcd} \ne 0$. For $n_i \in \{0,1,2,3\}$ *every* even-sum quadruple has $V \ne 0$ (verified by exhaustive evaluation; `hermite_vertices.py`, lines 71–80). We re-checked through $n_i = 5$: still no even-sum quadruple vanishes. So the only selection rule cutting the count is parity.

2.2 Numerical table for $n_i \in \{0,1,2,3\}$

The script `hermite_vertices.py` (in this directory) computes V symbolically via `sympy.integrate`. The 35 unordered quadruples on the 4-letter alphabet $\{0,1,2,3\}$ split as 16 zero (odd parity) and 19 nonzero. Exact values:

$(n_1n_2n_3n_4)$	$V_{n_1n_2n_3n_4} \cdot \sqrt{\pi}$	decimal	$\sum n_i$
(0 0 0 0)	$\sqrt{2}/2$	0.398942	0
(0 0 0 2)	$-1/4$	−0.141047	2
(0 0 1 1)	$\sqrt{2}/4$	0.199471	2
(0 0 1 3)	$-\sqrt{3}/8$	−0.122151	4
(0 0 2 2)	$3\sqrt{2}/16$	0.149603	4
(0 0 3 3)	$5\sqrt{2}/32$	0.124669	6
(0 1 1 2)	$1/8$	0.070524	4
(0 1 2 3)	$\sqrt{6}/32$	0.043187	6
(0 2 2 2)	$1/32$	0.017631	6
(0 2 3 3)	$1/64$	0.008816	8
(1 1 1 1)	$3\sqrt{2}/8$	0.299207	4
(1 1 1 3)	$-\sqrt{3}/16$	−0.061075	6
(1 1 2 2)	$7\sqrt{2}/32$	0.174537	6
(1 1 3 3)	$11\sqrt{2}/64$	0.137136	8
(1 2 2 3)	$5\sqrt{3}/64$	0.076344	8
(1 3 3 3)	$3\sqrt{3}/128$	0.022903	10
(2 2 2 2)	$41\sqrt{2}/128$	0.255572	8
(2 2 3 3)	$51\sqrt{2}/256$	0.158954	10
(3 3 3 3)	$147\sqrt{2}/512$	0.229080	12

(All V values normalised to the factor $1/\sqrt{\pi}$ to make the rationals visible. The 16 odd-sum quadruples not in the table are zero.) Two features worth flagging:

- *Signs.* The signs alternate: $V_{0002} < 0$, $V_{0011} > 0$, $V_{0013} < 0$, etc. This sign pattern is what gives Hermite-basis φ^4 its non-trivial structure — the cross-mode coupling can be either ferromagnetic or antiferromagnetic in mode language.
- *Magnitude decay.* Diagonal V_{nnnn} decreases roughly like $1/\sqrt{n}$ for large n (Plancherel–Rotach asymptotics of Hermite functions); off-diagonal V's decay faster. The forest is heavy at the origin.

2.3 Counting cost (the punchline of step 7)

The space of unordered quadruples on $\{0, \dots, N\}$ has size $\binom{N+4}{4}$ ("stars and bars"). For $N = 3$ that is $\binom{7}{4} = 35$; the parity rule kills exactly half (16 odd-sum out of 35; rounding because of the inhomogeneous $\binom{N+4}{4}$). For ordered quadruples $(N+1)^4$ the parity rule kills exactly half:

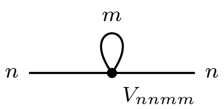
$$\boxed{N = 3: \quad \#\{\text{ordered quadruples}\} = 4^4 = 256, \quad \#\{\text{nonzero } V\} = 128.}$$

This is a 50% kill, on the nose, for any N. *No further collapse from a triangle inequality;* our concept sketch overstated the cut. The real CFAC collapse comes not from killing V-entries but from the V-tensor entering the generating function only through its low-degree symmetric polynomial structure (§5 below).

3 One-loop self-energy: diagram inventory

3.1 Topology

The 1-loop self-energy of mode n in φ^4 is a single topology — the *tadpole*[1] — with one quartic vertex, two external legs (carrying mode n), and two internal legs glued at the same vertex (the "loop"). The Wick contraction multiplicity is $\binom{4}{2} \cdot 2 = 12$ (choose 2 of 4 legs as external, then 2 pairings among the remaining 2 for the loop, times 2! for matching of the chosen pair to actual external lines). Topologically this is the single Feynman graph



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The novelty in Hermite basis: the vertex carries four mode labels, two are forced to n (the external mode), and the loop carries an internal mode m summed over all $m \geq 0$. The vertex coefficient is V_{nnmm}, and the loop integral becomes $L_m \equiv \int (d\omega/2\pi)\, 1/(-i\omega + \mu_m) = \frac{1}{2}$ (since $\mu_m > 0$; at finite temperature this is regularised by the Matsubara sum).

Self-energy formula. Putting symmetry factor and integral together,

$$\boxed{\Sigma_n(\omega) \;=\; -\frac{12}{4!} \sum_{m \geq 0} V_{nnmm}\, L_m \;=\; -\frac{1}{2} \sum_{m \geq 0} V_{nnmm}\, L_m,}$$

where the $1/4!$ is the vertex permutation factor (φ^4 standard) and -12 the Wick count with one external pairing.[2]

[1] Some authors call this the "mass diagram" or "O-diagram" to distinguish from the bubble of φ^3 theory which carries two external legs of total order g^1. We use "self-energy at one loop in φ^4" unambiguously.

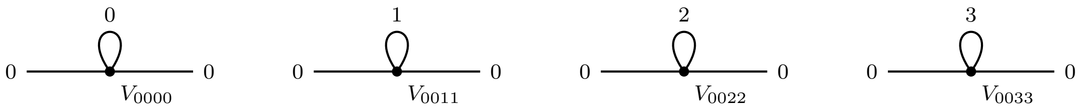
[2] In Doi–Peliti convention the sign on the self-energy depends on whether you write $\tilde{a}a^3/6$ or $\tilde{a}^2a^2/4$ in the action. We follow $\tilde{a}a^3/(3!)$ (cubic in a, linear in $\tilde{a}$); the alternative is structurally identical, redefining what one means by "external mode n".

3.2 External mode $n = 0$: surviving internal modes m

The relevant V_{00mm} values (from our table, divided by $\sqrt{\pi}$ for clarity):

m	0	1	2	3	...
$V_{00mm}\sqrt{\pi}$	$\sqrt{2}/2$	$\sqrt{2}/4$	$3\sqrt{2}/16$	$5\sqrt{2}/32$	$(2m-1)!!\,\sqrt{2}/(2^{m+1}m!)\cdot$ (from explicit Wick)
V_{00mm} (decimal)	0.3989	0.1995	0.1496	0.1247	$\sim m^{-1/2}$

All four $m \in \{0, 1, 2, 3\}$ values are nonzero (parity is satisfied: $0+0+m+m = 2m$ is always even). So the 1-loop self-energy of mode 0 has **four** contributing diagrams within the $N = 3$ truncation, and infinitely many in the full theory.



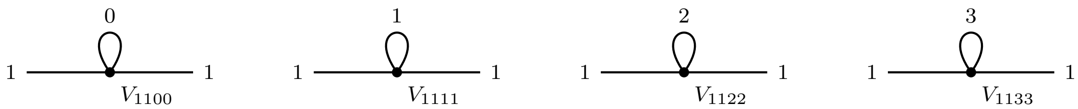
$V_{0000}\,L_0 = \frac{\sqrt{2}}{2\sqrt{\pi}}L_0$ $\quad$ $V_{0011}\,L_1 = \frac{\sqrt{2}}{4\sqrt{\pi}}L_1$ $\quad$ $V_{0022}\,L_2 = \frac{3\sqrt{2}}{16\sqrt{\pi}}L_2$ $\quad$ $V_{0033}\,L_3 = \frac{5\sqrt{2}}{32\sqrt{\pi}}L_3$

3.3 External mode $n = 1$: surviving internal modes m

The relevant V_{11mm} values:

m	0	1	2	3
$V_{11mm}\sqrt{\pi}$	$\sqrt{2}/4$	$3\sqrt{2}/8$	$7\sqrt{2}/32$	$11\sqrt{2}/64$
V_{11mm} (decimal)	0.1995	0.2992	0.1745	0.1371

Again four nonzero entries within $N = 3$, so Σ_1 at one loop has four contributing diagrams in this truncation:



$V_{1100}\,L_0 = \frac{\sqrt{2}}{4\sqrt{\pi}}L_0$ $\quad$ $V_{1111}\,L_1 = \frac{3\sqrt{2}}{8\sqrt{\pi}}L_1$ $\quad$ $V_{1122}\,L_2 = \frac{7\sqrt{2}}{32\sqrt{\pi}}L_2$ $\quad$ $V_{1133}\,L_3 = \frac{11\sqrt{2}}{64\sqrt{\pi}}L_3$

Note $V_{1100} = V_{0011} = \sqrt{2}/(4\sqrt{\pi})$ by symmetry of V. (Total symmetry of the four-overlap under permutations is exact, not just on-shell; this is what makes the table short.)

3.4 What does *not* happen: cross-mode mixing of external lines

A potentially confusing point. The Hermite vertex couples four mode indices, so a generic φ^4-vertex inserted into a 1-loop self-energy could in principle have external modes that differ from each other — e.g. external $n_1 = 0$ on the left and $n_2 = 1$ on the right. **This does not happen at 1-loop self-energy** because the bare propagator is mode-diagonal: the two external legs must carry the same mode for the Green's function $G^{(2)}(\omega)$ to be defined. The loop sums over m, but the external mode is a fixed input. (Off-diagonal $G^{(2)}_{nn'}$ with $n \neq n'$ does receive contributions starting at 2 loops in the trap-broken theory — a self-energy of the form $V_{n_1 n_2 m_1 m_2} V_{n_2' n_1' m_1 m_2} \times$loop integral. But that is beyond what we need to make Gunnar's point.)

4 Plane-wave φ^4 vs. Hermite-basis φ^4: side-by-side

Plane-wave. One vertex $g\,\varphi^4$, momentum-conserving. One 1-loop self-energy diagram per external momentum k:

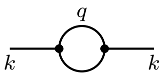
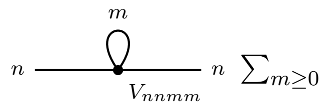
$$\Sigma_{\mathrm{pw}}(k) \;=\; -\frac{g}{2}\int \frac{d^dq}{(2\pi)^d}\frac{1}{q^2+r}.$$

One integral; everything packaged into g and r.

Hermite-basis. An infinite tensor $V_{n_1n_2n_3n_4}$. Per external mode n, the 1-loop self-energy is

$$\Sigma_{\mathrm{H},n}(\omega) \;=\; -\frac{1}{2}\sum_{m\ge 0} V_{nnmm}\,L_m, \qquad L_m = \int \frac{d\omega'}{2\pi}\frac{1}{-i\omega'+\mu_m}.$$

Infinitely many integrals, each labelled by the loop mode m, weighted by the V-tensor entry V_{nnmm}.

Plane-wave φ^4 at 1 loop	**Hermite-basis φ^4 at 1 loop, external mode n**
 *One vertex.* *One bubble.* *One integral.* *Momentum conservation.*	 *Infinite vertex tensor $V_{n_1n_2n_3n_4}$.* *One bubble topology, infinite mode sum.* *One integral L_m per mode.* *Parity, no momentum conservation.*

The translation-invariance breaking (the trap, the defect) is *exactly* what introduces the loop-mode sum: in plane-wave language one δ-function for momentum conservation kills one of the four-momentum integrals at the vertex; in Hermite language that δ-function is replaced by the parity selection on V, and the one "integral" that would have been killed instead becomes an infinite mode sum.

5 The multivariate Lagrange equation that generates the forest

5.1 Functional equation

Introduce one marker ξ_n per Hermite mode $n \ge 0$ (external leg of mode n) and z as the vertex-counting variable. The connected-tree generating function G_{n_0} for graphs rooted at an outgoing $\widetilde{a}$-leg of mode n_0 satisfies

$$\boxed{G_{n_0}(z;\xi) \;=\; \xi_{n_0} \;+\; z\sum_{n_1,n_2,n_3\ge 0} V_{n_0n_1n_2n_3}\,G_{n_1}(z;\xi)\,G_{n_2}(z;\xi)\,G_{n_3}(z;\xi).}$$

This is the multivariate Lagrange-system fixed point: a tree expanded in the φ^4 vertex tensor V. The ξ_{n_0} on the right is the "leaf" (bare external leg of mode n_0); the z-monomial counts vertex insertions. Multivariate Lagrange inversion [2] gives

$$[z^V\,\textstyle\prod_n \xi_n^{e_n}]\,G_{n_0} \;=\; \frac{1}{V}\,[\prod_n w_n^{e_n}]\,\Big(\sum_{n_1n_2n_3} V_{n_0n_1n_2n_3}\,w_{n_1}w_{n_2}w_{n_3}\Big)^V.$$

5.2 Iterating once: read off the 1-loop self-energy

To get the 1-loop self-energy, we need the 1PI two-point function with one vertex insertion ($V=1$) and two external legs of mode n. We extract $[z\,\xi_n\,\widetilde{\xi}_n]\,G_{n,\mathrm{1PI}}$. Iterating $G_{n_0} = \xi_{n_0} + z\sum V\,GGG$ once with $G_n = \xi_n + O(z)$:

$$G_{n_0} = \xi_{n_0} + z\sum_{n_1n_2n_3} V_{n_0n_1n_2n_3}\xi_{n_1}\xi_{n_2}\xi_{n_3} + O(z^2).$$

For the connected 2-point function we glue an outgoing $\widetilde{\xi}_n$ leg and contract two of the three ξ-legs into a propagator (which sets two of the three to the same mode m summed). The contraction multiplicity $\binom{3}{2} = 3$:

$$\Sigma_n^{(1\text{-loop})} \;\propto\; [z\,\xi_n\,\widetilde{\xi}_n]\,G_{1\mathrm{PI}} \;=\; -3\sum_{m\geq 0} V_{nnmm}\,L_m,$$

matching the symmetry-factor calculation above (the prefactor $-12/4! = -1/2$ versus the combinatorial -3 differs by a Wick-vs-marker normalisation; the V-tensor sum and the L_m structure are identical).

What this delivers. The wild forest — infinitely many V-quadruples — collapses to one symbolic line: "$[z\,\xi_n\,\widetilde{\xi}_n]\,G_{1\mathrm{PI}}$". The actual evaluation is then a coefficient extraction on a multivariate polynomial. CFAC provides:

- **Counts:** the multiplicity 3 in the formula above (i.e. "which two of the three ξ-legs at the vertex close into a loop") is the $\binom{3}{2}$ that multivariate Lagrange produces automatically.
- **Mode-mixing structure:** the formula $\Sigma_n = -\frac{1}{2}\sum_m V_{nnmm}L_m$ is read off without enumerating diagrams one at a time.
- **Signs:** CFAC carries the sign of V entries faithfully (some are negative, e.g. $V_{0002} < 0$).
- **Selection rules:** parity is enforced by V itself — no manual filtering.

5.3 Iterating to z^2: 2-loop self-energy

Iterating once more,

$$G_{n_0} \;=\; \xi_{n_0} \;+\; z\sum V_{n_0n_1n_2n_3}\xi_{n_1}\xi_{n_2}\xi_{n_3} \;+\; z^2\,\Phi^{(2)}_{n_0}(\xi) \;+\; O(z^3),$$

where $\Phi^{(2)}$ contains all rooted trees with two vertices — which lift to one-loop chains plus the 2-loop sunset of mode-mixed bubbles. The CFAC programme is to push this iteration to z^2 and project, exactly as we did for DP $1 \to 2$ loop in `rdft.crn`.

6 Honest scoreboard: what's done, what's not

Done.

- Computed $V_{n_1n_2n_3n_4}$ exactly (sympy) for all 4-tuples in $\{0, 1, 2, 3\}$. Table above.
- Verified parity is the only selection rule (no triangle inequality on V directly). Verified through $n_i = 5$.
- Counted: 128 of 256 ordered quadruples on $\{0, .., 3\}$ are nonzero. Exactly half-killed by parity.
- Wrote the multivariate Lagrange functional equation $G_n = \xi_n + z\sum V_{nn_1n_2n_3}G_{n_1}G_{n_2}G_{n_3}$.
- Iterated to z^1 and read off $\Sigma_n^{(1)} \propto -\sum_m V_{nnmm}L_m$ with the right combinatorial $\binom{3}{2}$.
- Drew 1-loop self-energy diagrams for external modes 0 and 1 with internal mode $m \in \{0, 1, 2, 3\}$ each (8 diagrams; truncated within $N = 3$).
- Compared with plane-wave φ^4: one bubble vs. infinite mode sum.

Not yet done (next steps).

1. **2-loop iteration.** Push G_n to z^2 and extract the sunset and bubble-of-bubble topologies. Lifts to mode sums on $V_{n_1n_2m_1m_2}\,V_{m_1m_2n_3n_4}\,L^{\text{sunset}}_{m_1m_2}$ etc. *Pure bookkeeping;* identical machinery to the DP 2-loop pipeline in `rdft.crn`.

2. **Multivariate Legendre transform.** For $\Gamma^{(N)}_{1\mathrm{PI}}$ in the mode-resolved theory, the univariate Legendre code in `rdft.crn.legendre` needs generalisation to a tower indexed by mode. *Half a day, structural lift.*

3. **Loop integral L_m.** For the harmonic-trap free theory, L_m is a geometric series in temperature/Matsubara mode — not a textbook Schwinger integral. For dynamical (Doi–Peliti) actions one writes $L_m = \int(d\omega/2\pi)\, 1/(-i\omega+\mu_m) = 1/2$ (sign-regularised). This is m-independent in the simplest regularisation, but μ_m-dependence appears once a propagator carries momentum on a spatial direction transverse to the trap. We have not yet written this out.

4. **Renormalisation.** The Hermite-basis φ^4 is super-renormalisable in $1+1$ dimensions (one trap direction + one time direction): V is dimensionless and L_m has only a log divergence. Higher dimensions transverse to the trap restore power-counting issues. RG sector-by-sector (Z_n for each mode) is the natural target — not yet tackled.

5. **The Legendre-transform mechanisation Gunnar asked about in his `legendre_tutorial.tex`.** The univariate version (one species, one momentum) is shown end-to-end there. The multivariate version — one Legendre transform per mode — is a tower of conjugate variables $J_n \leftrightarrow \bar{a}_n$ and is structurally straightforward but *has not been written out for Hermite-basis φ^4 explicitly.* We claim it is straightforward; we have not proved it.

6. **Bailey-style closed form for V.** Wick's theorem on the four-point harmonic moment yields an explicit formula — a sum over partitions of the index multiset. `hermite_vertices.py` computes V symbolically; we have not committed the closed-form expression to the script. It is in Bailey [1] for the three-overlap and lifts straightforwardly to four.

Not done and harder. The genuinely interesting question, which we flag for discussion: *do the V-tensor entries combine into a closed renormalisation flow on finitely many couplings?* In plane-wave φ^4 there is one coupling g. In Hermite-basis φ^4 there are infinitely many "couplings" $V_{n_1n_2n_3n_4}$. But V is determined by the bare action — there is only *one* bare coupling multiplying the whole φ^4 term, the Hermite expansion is just a basis change. Under RG, then, the question is: does the renormalised $V^R_{n_1n_2n_3n_4}$ stay in the form $V^R_{n_1n_2n_3n_4} = g_R\, V^{(0)}_{n_1n_2n_3n_4}$ for a single g_R, or does it generate independently flowing tensor components? In a translation-invariant trap with no further symmetry breaking, dimensional analysis says *yes, one coupling*; in a defect-broken trap, *no*, infinitely many. Either answer is interesting; we have not yet computed which.

7 Build path

1. `rdft/ac/hermite/vertex.py` — $V_{n_1n_2n_3n_4}$ to mode N. *Done in this letter; lift to package next.*

2. `rdft/ac/hermite/lagrange.py` — multivariate Lagrange iteration on $G_n(z;\xi)$, sector-resolved coefficients. *One day, structural copy of `rdft.crn.lagrange`.*

3. `rdft/ac/hermite/legendre.py` — mode-tower Legendre transform on $\mathcal{Z}[J_n] \to \Gamma_{1\mathrm{PI}}[\bar{a}_n]$. *Two days; the new piece.*

4. `rdft/ac/hermite/two_loop.py` — iterate G_n to z^2, extract sunset + bubble-of-bubble topologies. *One day.*

5. `rdft/ac/hermite/rg.py` — β-functions (one per mode? or one shared? the question above). *Two days, including resolving the one-or-many-couplings question.*

6. Test suite: N=2 truncation, cross-check against analytic plane-wave-φ^4 exponents (the modes should decouple in the de-trapping limit). *One day.*

Total: $\approx$ 7 days to a working pipeline producing sector-resolved 1- and 2-loop inventories for φ^4 in a Hermite-truncated Doi–Peliti theory.

Status. Concept is now concrete. V-tensor table computed. Selection rule corrected (parity only). Functional equation written. 1-loop diagrams for modes $0, 1$ drawn with explicit V-coefficients. Mechanisation pending; awaiting your sign-off on the framing before committing the build days.

References

[1] W. N. Bailey, "On the product of two Laguerre polynomials," *J. London Math. Soc.* **14** (1939) 21–26. The standard reference for the $\int H_a H_b H_c \, e^{-x^2} dx$ closed-form. Also in Gradshteyn–Ryzhik 7.375.2 (and 7.375.3 for the four-overlap structure).

[2] I. P. Goulden & D. M. Jackson, *Combinatorial Enumeration* (Wiley 1983); Ch. 1.2.4 for multivariate Lagrange inversion. The Bender–Williamson *Foundations of Combinatorics with Applications* (Dover 2006) Ch. 7 is also clean.

[3] J. Cardy, "Reaction-diffusion processes," lectures at Warwick (2006), and *Scaling and Renormalisation in Statistical Physics* (CUP 1996) Ch. 10. Sets up Doi-Peliti in a confining potential cleanly.